

GEORGE JOBI PERANGATTU

Robotics Software & Autonomous Systems Project Portfolio | gjobiper@asu.edu | (480) 742-9855 | linkedin.com/in/george-j-1829112a2/

TECHNICAL PROFILE

M.S. student in Robotics and Autonomous Systems (AI) at Arizona State University with a completed B.S. in Computer Science. Project work spans camera-LiDAR perception, state estimation, physical robot control, motion planning, simulation, sensor integration, testing, and manufacturing automation.

SELECTED PROJECTS

Efficient TransFuser: Camera-LiDAR Sensor Fusion for Localization & Perception

- Replaced the TransFuser backbone with EfficientNetV2-S, reducing parameters by 70% from 168M to 50M while improving held-out loss by 8.7%.
- Built multi-scale transformer fusion for RGB images and LiDAR point clouds, improving BEV segmentation by 25.4% across perception and localization tasks.
- Developed PyTorch data, training, and evaluation pipelines for detection, depth estimation, mapping, navigation, and segmentation in CARLA; diagnosed and resolved an FP16 focal-loss instability.

Parrot MiniDrone: Real-Time Localization & Autonomous Control

- Implemented closed-loop PID control for yaw, velocity, and altitude on physical hardware using inertial and visual sensor fusion for real-time state estimation.
- Built automated test sequences, sensor logging, performance evaluation, safety checks, and failsafes for repeatable autonomous operation.

3D Motion Planning for Autonomous Navigation in Gazebo

- Developed a Python 3D RRT planner integrated with ROS and Gazebo for collision-free path generation; evaluated and post-processed trajectories across multiple environments.

Spider CAD Robot & Factory Automation

- Designed a multi-legged robot with constrained joints, kinematic checks, actuator placement, and a URDF-ready path toward ROS/Gazebo validation.
- Preparing an industrial automation simulation integrating sensors, actuators, state tracking, and basic PLC/control logic for semiconductor-process workflows.

PRODUCTION SOFTWARE EVIDENCE

- DigiClips Media: built Python/PyTorch data and ML pipelines handling 10M+ daily queries, reducing latency by 45%, and improved computer-vision accuracy by 30%.
- Odoor and TicketDex: delivered production backend and full-stack improvements, including testing infrastructure and a 30% latency reduction.

Core tools: C/C++, Python, ROS/ROS2 concepts, Linux/Ubuntu, PyTorch, OpenCV, Gazebo, CARLA, Git, CMake, Docker, CI/CD, sensors, actuators, PID control, motion planning, computer vision, state estimation, and sensor fusion.